
Preston Rank

prestonrank5@gmail.com • 682-560-2759 • <https://github.com/vegettablee> • [linkedin.com/in/preston-rank](https://www.linkedin.com/in/preston-rank)

Education

University of Texas at Dallas

GPA: 3.57 | Bachelor of Science in Computer Science

Richardson, TX

Expected Grad: Dec 2027

Relevant Coursework: Software Engineering, Databases, Data Structures and Algorithms, Unix Programming

Experience

OrbitalMentorship — (Early-Stage Ed-tech Startup)

Richardson, TX

Software Engineer Intern

Aug 2025 - Jan 2026

- Increased mentor-mentee collaboration by shipping an **Express/React analytics dashboard** aggregating raw user events from the core onboarding and pairing module, converting engagement data into progress metrics that directly improved **feature prioritization** for **2+ stakeholders**.
- Gathered client requirements to resolve lack of centralized file storage, building an **Express/AWS S3** solution with **unit-testing** and retrieval validation that gave **50+ users reliable access** to resumes, assignments, and profiles.

ACM Research — (Capital of Texas Undergraduate Research Conference)

Richardson, TX | Austin, TX

Truncated SVD-Based Prompt Compression for LLMs

Aug 2025 – Mar 2026

- Collaborated with a small group to develop a prompt compression system, reducing **LLM prompt length by 40%** to **lower API costs** and improve response times while maintaining output quality.
- Presented research to a **judging panel** with varying technical backgrounds through a poster, including a separate **Q&A** portion that required defending technical decisions and explaining real world impact.

Starbucks | Nordstrom

Arlington, TX | Dallas, TX

Barista | Sales Associate

Jul 2022 - May 2025 | May 2025 - Present

- Trained and mentored **5+ new hires**, developing tailored training processes based on individual learning styles.
- Resolved customer concerns** during peak-volume AT&T events, de-escalating frustrated customers through clear communication during **30+ minute wait periods**.
- Consulted **high-value clients at NorthPark Center** with luxury purchases, explaining **technical details** (fabric, construction) accessibly tailored to each client's background, taste, and expectations.

HackTX 2025 — (Toyota Track)

Austin, TX

Car-ma! – Dating Inspired Car Recommendation App

Oct 2025

- Designed a car recommendation system that matched users to Toyota's inventory based on their needs, using vector embeddings and cosine similarity in **Spring Boot**, deployed on **DynamoDB** and **Vercel**.
- Collaborated cross-functionally with **frontend** and **UI/UX** teammates to align backend functionality with design requirements, delivering a smooth user friendly full stack experience.

Projects

iMessage RAG Query System with Conversation Disentanglement Model

Oct 2025 - Jan 2026

Github + Demo: <https://github.com/vegettablee/RAGiMessageFinder>

- Developed full **RAG** application in **Python** with an interactive **Chainlit** chat interface, and contact-based message extraction and processing from macOS iMessage database (**SQLite**).
- Built three-stage **message chunking pipeline** combining time-based segmentation, **ML model** conversation disentanglement (**74.1% F1 score** on IRC Disentanglement), and semantic similarity-based merging.

Internship Email Automation Agent Python CLI Tool

Dec 2025 – Mar 2026

- Built a **Dockerized** multi-agent CLI tool automating personalized internship outreach, integrating the Gmail API for end-to-end email delivery and tracking.
- Coordinated **Pydantic AI agents** for email personalization and resume selection using role and company-specific context, backed by a normalized **PostgreSQL database** with rate-limiting and caching to handle API constraints.

Mobile Thrift Store Discovery App

Jun 2025 – Aug 2025

- Built a thrift store discovery app that surfaces and categorizes stores beyond Google Maps, using **SwiftUI** and **MapKit** with direct filters for style, type, and budget to navigate **1000+ curated listings across Texas**.
- Automated store discovery and ingestion through **Node/Express**, multi-querying Google Places within the client's radius for GPT-5 metadata categorization, storing photos into **AWS S3** and **MongoDB** for client retrieval.

Technical Skills

- Languages:** Python, JavaScript, TypeScript, Bash, SQL, Java, Swift, C, C++
- Frameworks:** React, Express, Spring Boot, LangChain, SwiftUI, PyTorch, Hugging Face Transformers, FAISS
- Tools & Technologies:** AWS, Docker, Git, CI/CD, Vercel, PostgreSQL, DynamoDB, MongoDB, SQLite